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Water uptake by roots of maize and sunflower affects the radial transport of abscisic acid and its concentration in the xylem

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Abstract. The radial movement of *cis*-abscisic acid (ABA) has been investigated in young excised roots of *Zea mays* L. and *Helianthus annuus* L. which were grown hydroponically. In addition to the symplastic path, ABA was largely translocated across the root apoplast by solvent drag with the water in the transpiration stream. On the apoplastic path ABA may even cross the endodermis. Depending on the ABA concentration of the medium (range: 5–500 nM) and in the root apoplast, the solvent-drag component of the flow of ABA counteracted the dilution of ABA in the xylem caused by transpirational water flow. Acidification of the rhizosphere and of the root apoplast increased the apoplastic transport component. In sunflower, the apoplastic flow of ABA was significantly weaker than in maize roots. This was also indicated by the larger apparent reflection coefficient (σ_{ABA}) of sunflower roots for ABA (sunflower: $\sigma_{ABA} = 0.97 \pm 0.02$, n = 6 roots; maize: $\sigma_{ABA} = 0.68 \pm 0.06$, n = 6 roots; ±SD). For both species, σ_{ABA} was smaller than unity. Root reflection coefficients were affected by factors such as pH, ABA concentration of the medium, and by the suction force applied to excised root systems. Due to the complex composite structure of the permeation barrier in the root, the reflection coefficient estimated from solvent drag is also complex. Since unstirred layers affected the absolute value of the reflection coefficient, σ_{ABA} has been termed 'apparent'. It is concluded that the pH and ABA concentration of the soil solution as well as the transpiration rate (suction force) modify the intensity of the root-to-shoot signal which is influenced by an apoplastic bypass flow of ABA. The latter may be substantially affected by the existence of Casparian bands in the exodermis, which were lacking in the roots studied in this paper.

Key words: *Cis*-abscisic acid – *Helianthus* – Hydraulic conductivity – Reflection coefficient – Root – Water transport – *Zea*

Introduction

The role of *cis*-abscisic acid (ABA) as a long-distance stress signal that is released from the roots to the xylem seems to be well established, even if there are a few exceptions where conjugated ABA may act as a stress signal (Bano et al. 1993; Munns and Sharp 1994). However, there is some discussion of how the intensity of the ABA signal in the xylem develops. Jackson et al. (1996) and Else et al. (1994, 1995) pointed out that ABA, like other solutes, is diluted in the xylem when water flow is enhanced by increased transpiration. Similarly, computer simulations of Slovik et al. (1995) indicated substantial changes in ABA concentration in response to diurnal fluctuations of transpiration. Even when a cloud covers the sun for just a short moment, this would result in considerable fluctuations in the ABA concentration in the vicinity of guard cells. Slovik's calculations and Jackson's suggestions imply that the radial movement of ABA in roots takes place entirely in the symplast and that ABA is released into the xylem across plasma membranes of parenchyma cells around the vessels. Usually, it is thought that the release of ABA into the root xylem is independent of the intensity of radial water flow (J_V given in $m^3\ m^{-2}\ s^{-1}$), i.e. the concentration of ABA in the xylem will be given by $C_X^{ABA} = J_{ABA}/J_V$ (J_{ABA} = radial flow of ABA across the root in mol $m^{-2}\ s^{-1}$). Hence, an increase in transpiration will cause a dilution of ABA in the xylem.

Abbreviations and symbols: ABA = *cis*-abscisic acid; C_O^{ABA} = ABA concentration in the medium; C_X^{ABA} = ABA concentration in the xylem; J_{ABA} = ABA flow per unit root surface area; J_V = volume flow per unit root surface area; Lp_r = hydraulic conductivity of the root; P_S^{ABA} = permeability of the root for ABA; PTS = trisodium 3-hydroxy-5,8,10-pyrenetrisulfonate; σ_{ABA} = apparent reflection coefficient of the root for ABA

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Steudle (1994), on the other hand, has pointed out that, to some extent, solutes such as nutrient ions, may be transported apoplastically across the root cylinder. In the endodermis, the passage of solutes may be restricted to arrays, where Casparian bands are not yet mature (such as in the root tip or where root initials break across the endodermis) or may even be related to a permeability of Casparian bands. Apoplastically, ABA should be translocated with the transpiration stream by solvent drag. Thus, the dilution of solutes in the xylem vessels in response to increased transpiration could be compensated for. If there is a coupling between flows of water and ABA, the rate at which ABA is transported apoplastically into the xylem would depend on the intensity of apoplastic water flow. Abscisic acid has been shown to increase (Glinka 1980) or decrease (Markhart et al. 1979; Fiscus 1981) the hydraulic conductivity (Lp_r) of roots (Lp_r = per-unit-area hydraulic conductance in m s^{-1} MPa^{-1}). In our experimental system, ABA caused approximately a 100% increase in root Lp_r. Thus, the ABA signal should be weakened when dilution plays the dominating role under conditions of increased water flow. The flow of ABA through roots will depend on the concentration of ABA available in the soil solution and on the amount of ABA which is produced in the root symplast and released to the apoplast. Compared with the contribution of ABA produced in root tissue, the relative contribution of the ABA amount which is imported from the soil solution into xylem vessels plays an important role. Despite the low concentrations of ABA that occur within soil solution (1–10 nM), the effect of imported ABA could be substantial when the internal ABA production is low. Most of the ABA is synthesised in the cytosol of root cortical cells (Hartung and Davies 1993). Under conditions of a high endogenous ABA biosynthesis, the apoplastic transport component could be important provided that the ABA produced in root cells is delivered to the apoplast and then dragged into the xylem. The latter mechanism will be strongly affected by the pH of apoplast and symplast (Daeter et al. 1993; Slovik et al. 1995).

Data in the literature on radial ABA transport across roots are rare. Hartung and Behl (1975) have shown that radioactive ABA applied to stelar or cortical tissue of runner bean (*Phaseolus coccineus* L.) roots is readily translocated across the endodermis to the cortex and stele, respectively. However, from these data no conclusions about the transport path could be drawn. Fiscus (1982) and Markhart (1982) have used excised root systems of bean (*Phaseolus vulgaris* L.) to work out transport coefficients of ABA [permeability coefficients (P_S^{ABA}) and reflection coefficients (σ_s^{ABA})] from pressure-perfusion experiments. The P_S^{ABA} was rather high. Despite reflection coefficients close to unity, it was concluded that there was a substantial movement of ABA within the transpiration stream. *cis*-abscisic acid was applied to the root medium in micromolar concentrations. At the high water flow densities used, this should have resulted in very high concentrations of the hormone in the root caused by concentration polarisation effects at the permeation barrier (endodermis). In part, the use of non-physiological conditions was due to limitations in measuring low ABA concentrations in the xylem sap at that time.

In the present paper, we have used excised root systems of young sunflower and maize plants to measure flows of ABA and water across roots. Rather than applying pressure to the root medium, cut surfaces of excised roots were subjected to vacuum (subatmospheric or negative pressures) to induce water flows which simulate conditions in the intact transpiring plant. The aim of the study was to quantify the rates at which ABA is dragged into the root xylem with the water at different suction forces and how this would affect the concentration of ABA in root exudates. The ABA concentrations in the medium were between 5 and 500 nM, which includes the range of concentrations at which ABA is present in soil solution (0–10 nM; Hartung et al. 1996). From the results, we conclude that substantial amounts of ABA may be translocated with the water through an apoplastic route which bypasses protoplasts, even in the endodermis.

Materials and methods

Plant material. Seeds of maize (*Zea mays* L. cv. Garant FAO 240; Asgrow, Bruchsal, Germany) and sunflower (*Helianthus annuus* L.; Rollis, Hargelsberg, Austria) were germinated on filter paper soaked with 0.5 mM $CaSO_4$ for 4–6 d at 21 °C in the dark. Maize seedlings developed roots up to 100 mm long and primary leaves up to 30 mm long. Sunflower seedlings developed roots of up to 120 mm and shoots of up to 25 mm. Individual seedlings were transferred to aerated hydroponic culture vessels (100 ml). The nutrient solution contained the following nutrients (in mM) KH_2PO_4 1.5, KNO_3 2.0, $CaCl_2$ 1.0, $MgSO_4$ 1.0, and (in µM) FeNaEDTA 18, H_3BO_3 8.1, $MnCl_2$ 1.5 at a pH of 5.5. Plants were kept in a greenhouse with an additional light source (mercury vapour lamp; 200 µmol m^{-2} s^{-1}; day/night 16/8 h; 25/17 °C). Plants used in experiments were cultivated for 7 d (*Zea mays* L.) or for 6 d (*Helianthus annuus* L.). The roots of the maize seedlings used were 80–310 mm long. Root systems consisted of 5–10 seminal roots. The overall shoot length was 100–150 mm. Sunflower seedlings had a root length of 120–230 mm and a shoot length of 80–120 mm. Surface areas from the roots of maize and sunflower were between 0.013 and 0.0073 m^2 and between 0.0039 and 0.0098 m^2, respectively.

Root anatomy and root surface areas. Freehand cross-sections of primary roots of maize and sunflower plants were made at distances from the root tip of 20, 100, 200, and 300 mm. Sections were either stained for 1 h with 0.1% berberine hemisulfate and subsequently for 45 min with 0.5% toluidine blue O (w/v) (Brundrett et al. 1988) or were used without staining (autofluorescence). They were viewed under an epifluorescence microscope using a violet filter set (exciter filter 365 nm, dichroitic mirror FT 395, barrier filter LP 397 Zeiss, Oberkochen, Germany). Microscope pictures were documented using a video camera connected to a computer to produce colour pictures (Intas Colour LC 100C low light camera; Göttingen, Germany). Lengths of primary roots were measured with a ruler. Root surface areas were determined using an image analysing system based on a video camera and software (DIAS from Delta-T Device; Cambridge, UK). For better contrast, roots were stained for 30 s with 0.25 mM methyl violet. Root surface areas were calculated from projected areas of roots which were assumed to be cylindrical in shape. The system was calibrated with a special procedure for randomly oriented roots. During the experiments, calibration was continuously checked by measuring metal wires of known length and diameter. This minimised errors, particularly, when measuring roots with small diameters.

Measurement of xylem sap flow of maize and sunflower roots at low subatmospheric pressures induced by vacuum. Seedlings were cut 20 mm below the root base. Excised roots with the mesocotyl (maize) and the hypocotyl (sunflower) still attached to it, were fixed to a capillary using a pressure-tight silicone seal fixed by a screw (Fig. 1). Silicone seals were individually prepared for each plant. They had a length of 15 mm and a hole individually adapted to the diameter of the mesocotyl or hypocotyl. The root medium was aerated. Suction applied to the root system caused xylem sap flow into a calibrated capillary. After 15 min at −0.02 MPa applied to the capillary, the flow of water across the root system was steady. After this period of time, xylem sap was collected. Water flow was determined by weighing harvested xylem sap fractions. It should be noted that the atmosphere was used as a reference (zero pressure) throughout in this paper. The suction forces applied to the roots have, therefore, a negative sign. This convention to use atmospheric pressure as the reference is common in most of the botanical literature.

Transport of ABA across roots of maize and sunflower and root hydraulic conductivity. At an applied pressure of −0.03 MPa, xylem sap was collected from the capillary in 20-min intervals. After 100 min, the nutrient solution was exchanged for a solution containing the same nutrients plus 100 nM ABA. Collection of sap was continued for a further 100 min. During suction experiments, J_v and C_X^{ABA} and, hence, J_{ABA} were steady after about 10 min. Root hydraulic conductivity (Lp_r in m s^{-1} MPa^{-1}) was estimated as the quotient of J_V and the used pressure difference of 0.03 MPa.

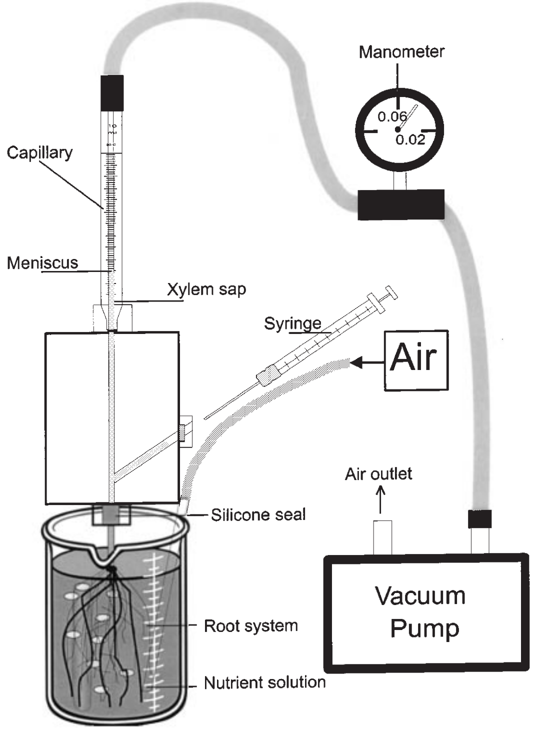

Fig. 1. Experimental set-up used to measure xylem sap flow under negative pressure applied to the xylem of excised root systems of young maize and sunflower plants. Roots were tightly fixed to a capillary by silicone seals using a screw. Subatmospheric pressures ranging from −0.02 MPa to −0.06 MPa were created with the aid of a pump, thus, imitating the situation in a transpiring plant

Flow of ABA across roots of maize and sunflower at different subatmospheric pressures (−0.02 and −0.06 MPa). Two hours before the experiments were started, plants were incubated in nutrient solution containing ABA at concentrations ranging from 5 to 500 nM. For 1 h, xylem sap was collected at intervals of 10 min at −0.02 MPa. Then subatmospheric pressure was decreased to −0.06 MPa and sap collection continued for another hour.

Determination of the apparent reflection coefficient, σ_{ABA} of roots for ABA. The reflection coefficient of ABA was calculated from solvent drag using Eq. 1 (Steudle 1994):

$$J_{ABA} = P_S^{ABA} \cdot (C_O^{ABA} - C_X^{ABA}) + (1 - \sigma_{ABA}) \cdot 0.5 \cdot (C_O^{ABA} + C_X^{ABA}) \cdot J_V + J^*_{ABA}. \quad \text{(Eq. 1)}$$

According to Eq. 1, the uptake of water (J_V) and of ABA (J_{ABA}) would be denoted as positive flows; P_S^{ABA} is the permeability coefficient of the root for ABA and C_O^{ABA} the concentration of ABA in the medium. The term $P_S^{ABA} \cdot (C_O^{ABA} - C_X^{ABA})$ represents the passive diffusion of ABA across the root cylinder according to Fick's first law and $(1 - \sigma_{ABA}) \cdot 0.5 \cdot (C_O^{ABA} + C_X^{ABA}) \cdot J_V$ the solvent-drag component of the overall flow of ABA. The term J^*_{ABA} denotes the active ABA transport across the root. As discussed later, terms for active ABA transport and diffusion can be neglected (see *Discussion*). Thus, Eq. 1 reduces to Eq. 2:

$$J_{ABA} = (1 - \sigma_{ABA}) \cdot 0.5 \cdot (C_O^{ABA} + C_X^{ABA}) \cdot J_V. \quad \text{(Eq. 2)}$$

Steady-state J_{ABA} was calculated from $J_{ABA} = C_X^{ABA} \cdot J_V$. Introducing this into equation (2) yields:

$$\sigma_{ABA} = 1 - \frac{2C_X^{ABA}}{C_X^{ABA} + C_O^{ABA}}. \quad \text{(Eq. 3)}$$

The term σ_{ABA} should be denoted as an 'apparent' root reflection coefficient, because the estimation of the concentration of the solute (ABA) within the permeation barrier of the root (endodermis) as the arithmetic mean of the external (medium) concentration and that of the xylem assumes a single homogeneous barrier. However, because of the composite structure of the root, this may not be a good approximation of the real situation (see *Discussion*). There may be problems with an accumulation of the solute in front of, or within the barrier (unstirred layers). Hence, the arithmetic mean may represent a lower limit of the actual concentration. In this case, the σ_{ABA} calculated from Eq. 3 would be a lower limit as well. Depending on the absolute value of the external ABA concentration, accumulation in the barrier will also tend to increase the solvent-drag component of J_{ABA} (Eq. 2).

Measurement of the apparent reflection coefficient for ABA of maize and sunflower root systems at different external pH. Two hours before starting the experiment, plants were incubated in nutrient solution containing 100 nM ABA and different buffers, 10 mM phosphate or Hepes-KOH, to establish a pH of 4.8 or 8, respectively. The unbuffered nutrient solution had a pH of 5.5.

Apparent reflection coefficient of severed roots. Roots of maize and sunflower plants were preincubated for 2 h in a nutrient solution containing 100 nM ABA. At a subatmospheric pressure of −0.03 MPa, xylem sap samples were taken every 10 min for 1 h. Subsequently, the tips of two primary roots of the corn root system, and one tip of sunflower, were severed at a distance of 60 mm from the apex. After cutting, sampling was continued for another hour.

Analysis of +ABA. The xylem sap samples were taken up with Tris-buffered saline buffer (Tris-HCl: 50 mM Tris, 150 mM NaCl, 1 mM $MgCl_2$, pH 7.8) and analysed immunologically by an enzyme-linked immunosorbent assay (ELISA) as described by Weiler (1986). Purification of samples was not necessary.

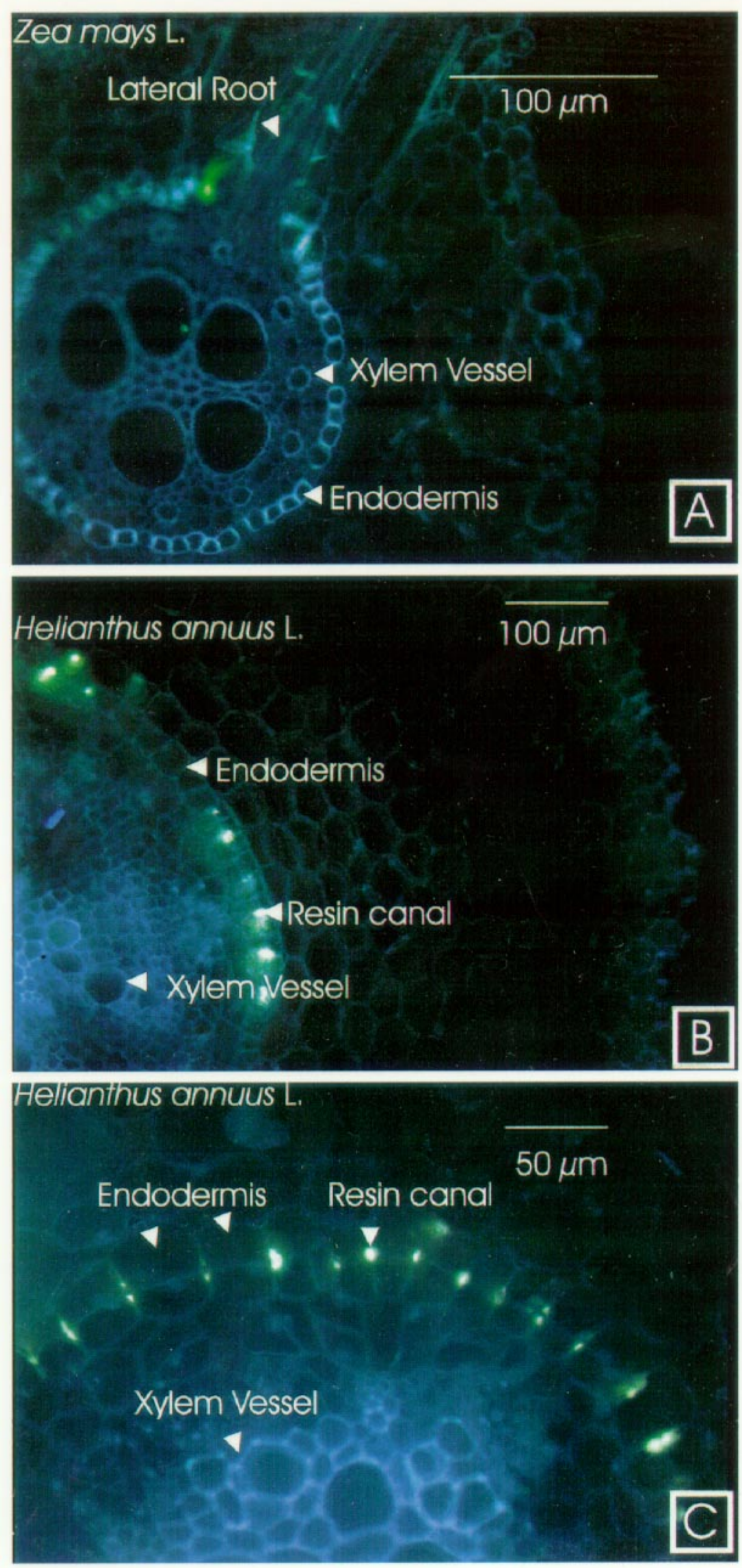

Fig. 2A–C. Freehand cross-sections of main roots of 7-d-old maize **(A)** and of 6-d-old sunflower plants (**B, C**). Maize roots (**A**) were stained with berberine hemisulfate, counter-stained with toluidine blue O, and viewed with an epifluorescence microscope (ultraviolet excitation; wavelength 365 nm). Staining with berberine hemisulfate caused intensive fluorescence of mature xylem, Casparian bands and suberin deposits. Counterstaining with toluidine blue O reduced berberine-induced fluorescence of lignin and suberin, but not of Casparian bands. In maize, endodermal Casparian bands are seen as light strips in the radial walls. In the endodermal layer of unstained sunflower roots (**B, C**), autofluorescence of Casparian bands could not be observed, xylem vessels were visible. Fluorescent points in the cell wall layer behind the expected endodermal layer are canals for resin or fatty waste products of metabolism. In maize, the cross-section was taken at a distance of 110 mm from the root tip behind the root hair zone and in sunflower, 20 mm behind the root base

Results

Anatomy. Maize roots grown in hydroponic culture developed a primary endodermis with a Casparian band in the radial cell walls at distances between 25 and 30 mm from the root tip. At larger distances of 60–100 mm, the endodermis became secondary and fluorescent suberin deposits could be observed in the cell walls of the endodermis. Between 180 and 250 mm, U-shaped wall thickenings occurred which are typical of a tertiary endodermis. At distances larger than 100 mm, lateral roots were visible. Their breaking through the endodermis (Fig. 2A, arrowhead) should have caused transient apoplastic bypasses (Steudle et al. 1993; Frensch et al. 1996). Maize roots from hydroponics never developed a continuous exodermis (hypodermis with a Casparian band). However, occasionally, patches of cells with suberised walls were found. It appeared that they developed in response to injuries or where laterals passed across the cortex and epidermis (Zimmermann and Steudle 1998).

In hydroponics, sunflower roots developed neither a suberised endodermis nor exodermal Casparian bands that were visible under the light microscope or stainable with berberine hemisulfate (Fig. 2B,C) for observation by fluorescence microscopy. Autofluorescent spots in the cell wall layer behind the endodermal layer of unstained cross-sections were canals for resin or fatty waste products of metabolism (Fig. 2B,C, arrowheads; Tetley 1925; Williams 1954). These spots could be removed by rinsing the sections with methanol/chloroform (1/1; v/v).

Radial flow of ABA. At a subatmospheric pressure of −0.03 MPa applied to the xylem, the ABA concentration of xylem sap exuded from excised sunflower roots ranged between 0.5 and 5 nM. In corresponding exudates from maize, concentrations were between 2 and 13 nM (Fig. 3A). After addition of 100 nM ABA to the medium, ABA concentrations in the xylem sap of *Zea mays* L. increased to values of up to 22 nM (22% of external ABA) at a pressure difference of 30 kPa. By contrast, the ABA concentration in the xylem sap of sunflower did not increase much under the same conditions (1.6–6.3 nM ABA; Fig. 3A). Hence, when the ABA concentration of the medium was increased to 100 nM, the corresponding increase of J_{ABA} was larger by a factor of 5.6 in root systems of maize than in those of sunflower (Fig. 3C). This was the case, although the addition of ABA increased water flow by 68% in maize and by 127% in sunflower (Fig. 3B). After adding 100 nM ABA to the medium, Lp_r values were significantly larger for both species (student *t*-test; $P \leq 0.05$; $n = 4$). On average, Lp_r increased from 1.3×10^{-7} to 2.2×10^{-7} m s^{-1} MPa^{-1} for maize and from 6.6×10^{-8} to 1.6×10^{-7} m s^{-1} MPa^{-1} for sunflower. Absolute values of Lp_r were similar to those given in the literature that were measured using root pressure probes and pressure perfusion techniques (e.g. Zhu and Steudle 1991; Zimmermann and Steudle 1998).

Measurement of ABA flow at different subatmospheric pressures. When a low suction force of −0.02 MPa was applied to the xylem, roots of sunflower and maize incubated in a nutrient solution containing 500 nM ABA attained steady ABA concentrations in the xylem sap of 6–19 nM and 6–22 nM, respectively. The ABA flows calculated from C_X^{ABA} and J_V were, on average, 8.3×10^{-14} mol m^{-2} s^{-1} and 5.6×10^{-14} mol m^{-2} s^{-1}, respectively. For both species, C_X^{ABA} was first measured during a period of 60 min at a subatmospheric pressure of −0.02 MPa. After decreasing the pressure to −0.06 MPa, C_X^{ABA} increased up to 40 nM in sunflower and up to 70 nM in maize. Therefore, J_{ABA} increased by a factor of 3.5 in sunflower and by a factor of 8 in maize. The radial flow of ABA across the root was calculated using the rates of increase of C_X^{ABA} and of J_V that were referred to unit area of root surface at both −0.06 MPa and −0.02 MPa (Fig. 4). In both species, J_{ABA} increased with increasing suction force. Following a step change in suction force from −0.02 to −0.06 MPa, ABA flow was larger by a factor of 2.3 in maize than in sunflower. In Table 1, ratios of $J_{ABA}(-0.06\ \text{MPa})/J_{ABA}(-0.02\ \text{MPa})$ are summarised for maize and sunflower at external ABA concentrations between 5 nM and 500 nM. The data show that for all measurements, the above-mentioned ratio was ≥1. A dilution of the ABA concentration in the xylem sap in response to the increased water flow at higher tension was never observed. On the contrary, the higher suction force caused J_{ABA} to increase more than J_V.

Determination of the apparent reflection coefficient of ABA (σ_{ABA}) at different external ABA concentrations. Equation 3 was used to calculate σ_{ABA} for maize and sunflower roots at different external ABA concentrations (Table 2). At 10 nM of external ABA, σ_{ABA} of maize was rather low ($\sigma_{ABA} = 0.4 \pm 0.07$; $n = 6$ at −0.02 MPa and 0.28 ± 0.05; $n = 6$ at −0.06 MPa), but increased to average values of between 0.6 and 0.9, when the ABA concentration increased from 20 to 500 nM. The apparent reflection coefficient was consistently lower at high pressure differences (0.06 MPa) than at low pressure differences (0.02 MPa). For sunflower roots, σ_{ABA} was 0.5 at 5 nM of external ABA and increased to values close to 1 in the presence of 100 nM ABA.

In Fig. 5 we show that σ_{ABA} of sunflower roots remained at values close to 1, when water flow was increased in the presence of a higher suction force. For maize roots, however, σ_{ABA} slightly decreased under these conditions.

The effect of external pH on σ_{ABA} of sunflower and maize roots (Figs. 6, 7). When the pH of the root medium was lowered to the pK_a of ABA of 4.8, σ_{ABA} was reduced significantly in both species (student *t*-test; $P \leq 0.05$; $n = 6$). The effect was more pronounced in maize than in sunflower and at higher pressure differences. However, under more acidic conditions (pH 3), no steady-state value of σ_{ABA} was attained within 1 h (data not shown). Apparently, this was due to a loss of selective properties by the roots.

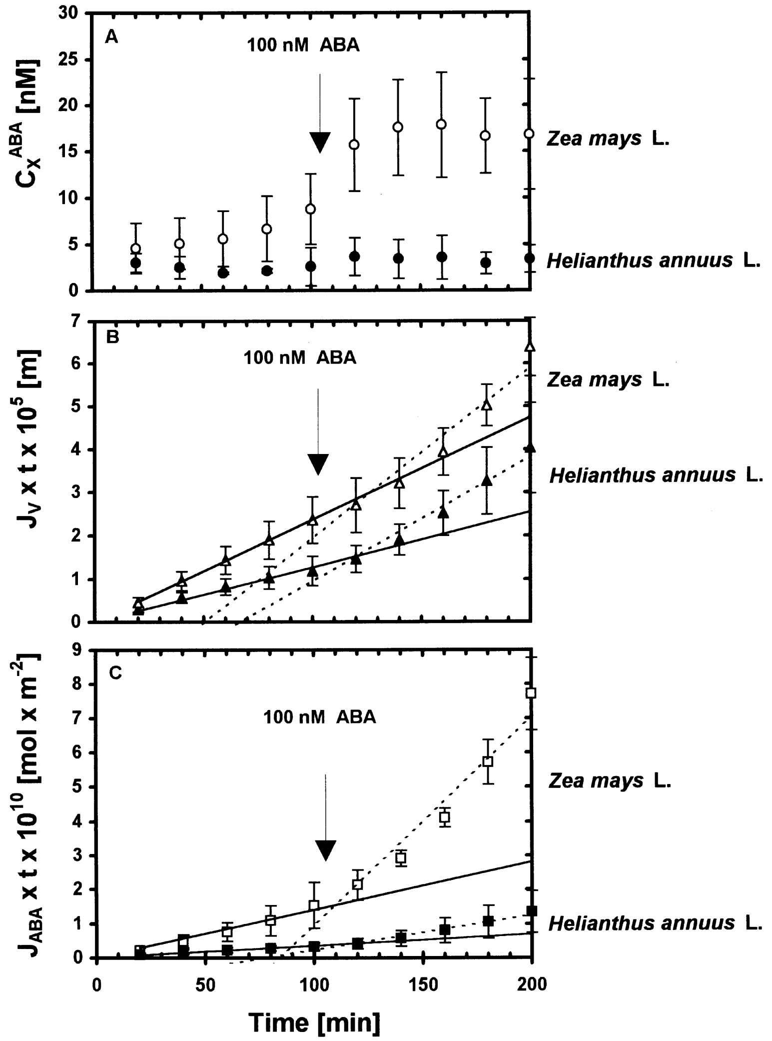

Fig. 3A–C. Xylem ABA concentration (●, **A**), water flow (▲, **B**), and radial flow of ABA (■, **C**), through roots of maize (*open symbols*) and sunflower (*filled symbols*) at a negative pressure of −0.03 MPa before and after addition of 100 nM ABA to the root medium (*arrow*). Data are means ± SD (n = 3 roots)

Determination of the reflection coefficient after severing tips of maize and sunflower roots. When root systems were severed by removing tips 60 mm behind the apex, there was a decrease in σ_{ABA} for both species (Table 3). This decrease was not significant in maize but was significant in sunflower (student t-test; $P \leq 0.05$; $n = 8$). After cutting, J_v increased by factors of 2.0 (maize) and 1.8 (sunflower) on average (Table 3).

Discussion

The data presented in this paper show that the ABA concentration in the xylem is not diluted when water flow across the roots is increased in the presence of external ABA in the concentration range 5–500 nM. For the low-nanomolar concentration range our experimental conditions are comparable to those present under natural conditions. In soil solution under various crops, ABA concentrations of 1–10 nM have been found, namely at low soil pH values (Hartung et al. 1996). The observation that an uptake of ABA from the medium by solvent drag may compensate for the dilution of ABA in the xylem by water uptake does not agree with the suggestions of Jackson et al. (1996) and Else et al. (1994, 1995). If this turns out to be generally true, the conventional assumption that the ABA concentration is diluted by increased water uptake, would have to be reconsidered. Uptake of ABA by solvent drag would have to be taken into account as well. Transport of ABA across roots may differ from that of nutrient ions. For nutrients, it has been shown that water and solute flows are largely uncoupled in roots (Anderson 1976; Miller 1985).

In part, our finding of a substantial solvent drag of ABA is in line with the results of Fiscus (1982) and of Markhart (1982) who measured transport coefficients of ABA in bean root systems. These authors pressurised excised root systems that were sitting in nutrient solution containing ABA at micromolar concentrations. They found a high permeability coefficient of ABA for the roots ($P_S^{ABA} = 3.5 \times 10^{-9}\,\mathrm{m\,s^{-1}}$) at a σ_{ABA} of close to unity. The latter result (high σ_{ABA}) is similar to our finding for sunflower but disagrees with the result from maize where σ_{ABA} was substantially smaller than unity. Thus, there may be considerable differences between species. The transport coefficients reported by Fiscus (1982) and by Markhart (1982) were obtained in the

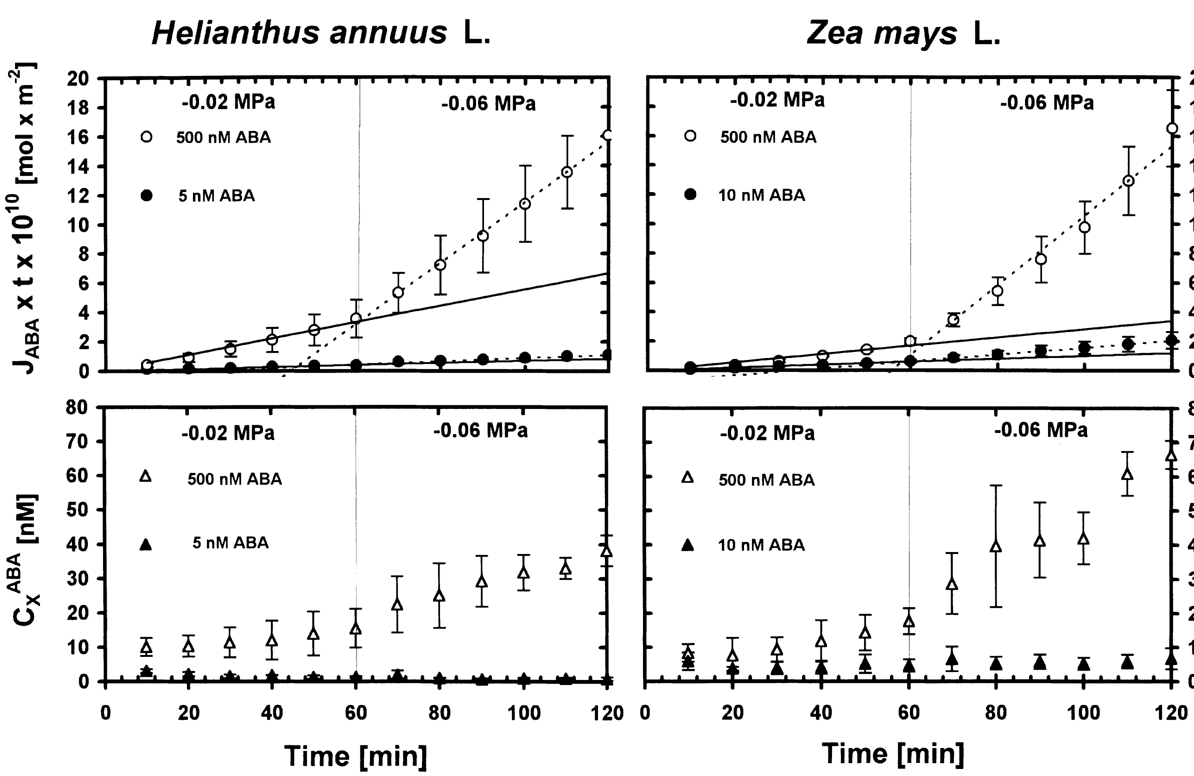

Fig. 4. Effect of different subatmospheric pressures (−0.02 MPa and −0.06 MPa) on the flow (J_{ABA}, -●-) and on the xylem concentration of ABA (C_x^{ABA}, ▲) in roots of maize and sunflower at different externally applied ABA concentrations (500 nM: *open symbols*; 5 nM and 10 nM: *filled symbols*). Data are means ± SD (n = 3 roots)

presence of external ABA concentrations which were larger by 2–3 orders of magnitude than in the present study. They also refer to large volume flows. This should have caused large ABA concentrations in the root apoplast. Despite the high reflection coefficient, the ABA flow density in the experiments of Fiscus (1982) and of Markhart (1982) was larger by 3 orders of magnitude than those measured in this paper. Therefore, Fiscus and Markhart could not completely exclude a substantial contribution of the diffusional and/or active component of ABA flow. However, since J_{ABA} increased with increasing water flow, they favour (despite $\sigma_{ABA} \approx 1$) the idea that ABA is dragged into the xylem with transpiration stream. In our experiments, the external ABA concentration was much smaller, which should have reduced the diffusional component drastically (see below). At an external ABA concentration of 10 μM, Fiscus (1982) found a typical dilution of the C_X^{ABA} value. However, this may be due to the fact that under these conditions (high external ABA concentration and water flow), the dilution effect was overriding the solvent-drag component which may be rather effective at the physiological concentrations used in this study.

A simple dilution effect for the ABA in the xylem was included in the computer models of Slovik et al. (1995), who simulated the long-distance transport of ABA in intact plants. Slovik's model was based on the assumption that radial ABA transport occurs exclusively in the symplast. The release of ABA from the symplast of xylem parenchyma cells into the lumena of vessels was

Table 1. Pressure-perfusion experiments with excised root systems of maize and sunflower at different external ABA concentrations. The ratios (± SD) of ABA (J_{ABA}) and water (J_V) flows at high (0.06 MPa) and low (0.02 MPa) pressure differences applied to cut shoots (Fig. 1) are given. Except for sunflower at low concentrations, there was a stronger increase in J_{ABA} than in J_V as suction force (water flow) increased. –, not measured

C_O^{ABA}	*Zea mays* L.		*Helianthus annuus* L.	
(nM)	Ratio of $J_{ABA(-0.06\ MPa)}/J_{ABA(-0.02\ MPa)}$	Ratio of $J_{V(-0.06\ MPa)}/J_{V(-0.02\ MPa)}$	Ratio of $J_{ABA(-0.06\ MPa)}/J_{ABA(-0.02\ MPa)}$	Ratio of $J_{V(-0.06\ MPa)}/J_{V(-0.02\ MPa)}$
500	5.9 ± 1.8 n = 5	1.9 ± 0.3 n = 5	3.5 ± 1.2 n = 5	1.7 ± 0.2 n = 5
100	3.4 ± 1.1 n = 5	2.0 ± 0.4 n = 7	2.0 ± 0.7 n = 6	1.7 ± 0.3 n = 6
70	3.6 ± 1.5 n = 4	1.6 ± 0.3 n = 6	–	–
50	3.4 ± 1.4 n = 8	1.7 ± 0.3 n = 12	–	–
20	2.5 ± 0.3 n = 5	2.1 ± 0.5 n = 5	–	–
10	2.6 ± 1.1 n = 6	1.7 ± 0.3 n = 6	–	–
5	3.5 ± 1.2 n = 4	1.7 ± 0.3 n = 5	1.1 ± 0.3 n = 5	2.1 ± 0.4 n = 7

Table 2. Apparent reflection coefficient of abscisic acid (σ_{ABA} ± SD) for young root systems of maize and sunflower. It can be seen that σ_{ABA} decreased as the pressure difference (or volume flow) increased. On the other hand, an increase in the ABA concentration of the root medium (C_O^{ABA}) caused an increase in the root reflection coefficient for the hormone. For further explanation, see text. –, not measured

C_O^{ABA} (nM)	Pressure difference	Apparent reflection coefficient σ_{ABA} of *Zea mays* L. $n = 6$	*Helianthus annuus* L. $n = 6$
500	0.02 MPa	0.96 ± 0.02	0.94 ± 0.01
	0.06 MPa	0.83 ± 0.05	0.87 ± 0.03
100	0.02 MPa	0.68 ± 0.06	0.97 ± 0.02
	0.06 MPa	0.54 ± 0.04	0.97 ± 0.02
50	0.02 MPa	0.71 ± 0.04	–
	0.06 MPa	0.48 ± 0.09	–
20	0.02 MPa	0.66 ± 0.05	–
	0.06 MPa	0.51 ± 0.09	–
10	0.02 MPa	0.4 ± 0.07	–
	0.06 MPa	0.28 ± 0.05	–
5	0.02 MPa	0.08 ± 0.15	0.49 ± 0.18
	0.06 MPa	−0.1 ± 0.10	0.68 ± 0.10

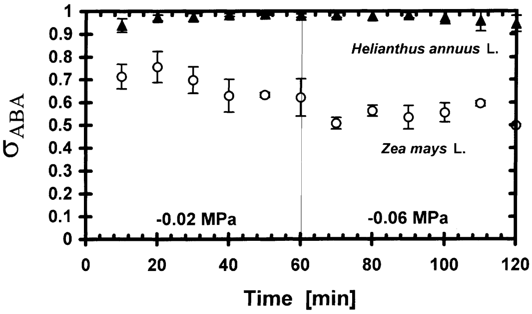

Fig. 5. Apparent reflection coefficient (σ_{ABA}) of sunflower (▲) and maize (○) roots as a function of harvest time. External ABA concentration was 100 nM; pH of the medium was 5.5. Data are means ± SD (n = 3–4 roots)

the rate-limiting step. Therefore, water flow caused a reduction of C_X^{ABA}. However, Steudle (1994) has postulated that solutes may be translocated laterally by solvent drag across an apoplastic bypass directly to the xylem vessels. At least to some extent and depending on the absolute values of root reflection coefficients, the apoplastic ABA concentration and the rate of water flow should counteract dilution.

A prerequisite for the evaluation of J_{ABA} and σ_{ABA} in our experiments was the condition of stationary flows of water and ABA. Taking a total volume of xylem vessels of 8–15 µl and a volume of water of 10–30 µl that is transported laterally to the xylem within 15 min, we assume that the total xylem sap of the root system is exchanged. Root Lp_r and σ_{ABA} were given as an average value for entire root systems, although it is known that transport coefficients and flows change along developing roots (Melchior and Steudle 1993; Frensch et al. 1996). Also, in the root systems used, the late metaxylem would have matured in the basal parts and suberin lamellae would have formed here, too. The very tips of the roots were free of mature xylem and therefore hydraulically isolated from the rest of the root (Steudle 1994). However, these complications are, perhaps, not as important in the context of this paper, since the hydraulic conductivity of the maize root systems estimated in this paper (mean $Lp_r = 1.3 \times 10^{-7}$ m s^{-1} MPa^{-1}) was similar to those for individual roots (Steudle and Frensch 1989; Zhu and Steudle 1991). Hence, effects of root development were probably not too big. For sunflower (mean $Lp_r = 6.6 \times 10^{-8}$ m s^{-1} MPa^{-1}), no data on hydraulic conductivity have yet been published. In our experimental system, apoplastic bypass flow was induced solely by applying hydrostatic pressure gradients, i.e. the water potential was reduced by vacuum applied to the excised xylem.

Different authors have used PTS (trisodium 3-hydroxy-5,8,10-pyrenetrisulfonate) as a tracer for apoplastic water, assuming that this solute is as mobile as water in cell walls (Hanson et al. 1985; Moon et al. 1986; Skinner and Radin 1994). Zimmermann and Steudle (1998), however, concluded that PTS dragged along with the water in the apoplast to the xylem does not trace the flow of water within the root apoplast. During pressure-perfusion experiments with different species, PTS flows were quite low. Only 0.01% to 0.6% of the PTS applied

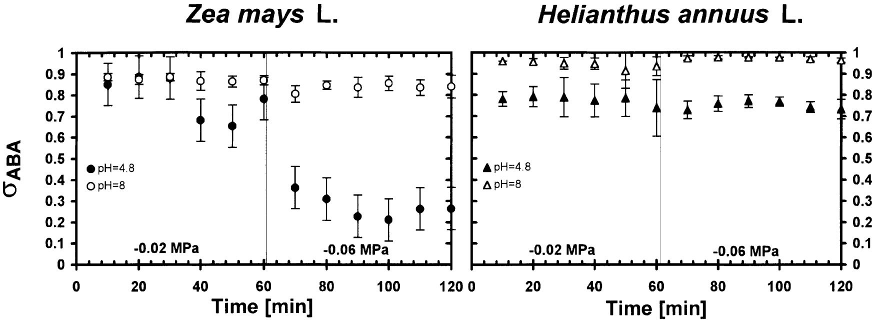

Fig. 6. Time course of the effect of high (pH = 8; *open symbols*) and low pH (pH = 4.8; *filled symbols*) on the σ_{ABA} of maize (●) and sunflower (▲) roots. It can be seen that there was a rapid decline of σ_{ABA} in maize but not in sunflower. During the experiments, the external ABA concentration was 100 nM. Data are means ± SD (n = 3–4 roots)

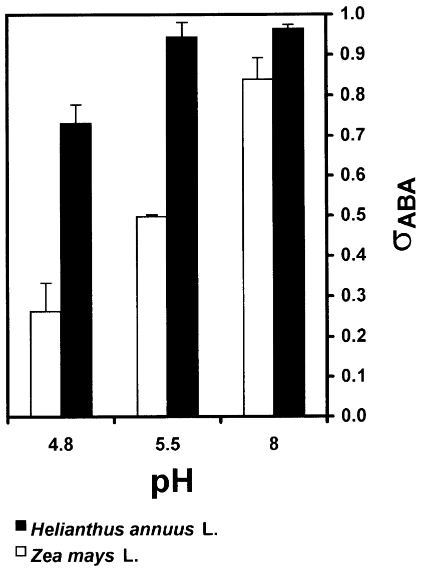

Fig. 7. Steady effects of external pH (pH = 4.8, 5.5 and 8) on the σ_{ABA} of maize (□) and sunflower (■) roots. It can be seen that for both species, σ_{ABA} increased with increasing pH. External ABA concentration was 100 nM. Data are means ± SD (*n* = 3–6 roots)

outside was found in the xylem sap. The result agrees with unpublished data of Hartung and Gloy who used PTS on maize and sunflower. In the present experiments, bypass flow of ABA caused a xylem concentration which was as large as 10% (maize) and about 5% (sunflower) of that of the medium, when the nutrient solution contained 500 nM of the solute. The differences may be explained by the fact that the molar weight of PTS (524 Da) is larger by a factor of 2 than that of ABA (264 Da). In addition, PTS has three sulfonic acid groups which are dissociated at physiological pH. Hence, PTS moves as an anion across the porous apoplast which bears negative fixed charges. It is, therefore, not surprising that apoplastic transport of PTS is restricted. According to Zimmermann and Steudle (1998) this throws some doubt on the use of PTS (and of other dyes) as a tracer for apoplastic water.

The situation described in the literature for PTS is different from that for ABA. As a monobasic carboxylic acid, ABA is less polar than PTS. The pK_a of ABA is 4.8 so that a considerable fraction will be not dissociated in the apoplast at regular pH. In the protonated form, ABAH moves across plasma membranes from the symplast into the apoplast. Ratios of concentrations in the two compartments depend on the pH values. To some extent, ABA will be trapped in the compartment of higher pH (Slovik et al. 1995). So, for different reasons it is not surprising that the apoplastic bypass flow of ABA can be much larger than that of PTS.

When radial water flow was increased, C_X^{ABA} was never diluted (Table 2). In maize, C_X^{ABA} even increased with increasing J_V. From the differences between maize and sunflower, we conclude that the barrier to ABA permeation in sunflower is much more restrictive than that in maize. Accordingly, the ‘passive selectivity’ as expressed by the reflection coefficient was larger in sunflower than in maize. As a consequence, more ABA molecules are ‘reflected’ from the barrier in sunflower than they are in maize when ABA is dragged across the root cylinder with the water. For young roots grown hydroponically, the permeation barrier would have to be largely identified by the endodermis (Steudle et al. 1993; Steudle and Peterson 1998). This, however, may be different for plants grown in soil or in aeroponic culture. Under these conditions, the exodermis could be a barrier

Table 3. Effect of severing individual roots of root systems of maize and sunflower on the overall reflection coefficient of ABA (apparent reflection coefficient, σ_{ABA}. Effects were smaller for maize than for sunflower root systems. The latter were less permeable to ABA and had a significantly higher root reflection coefficient as calculated from solvent drag

Root no.	σ_{ABA} *Zea mays* L.			
	Intact root tip	Root tip severed	σ_{ABA} intact/ σ_{ABA} severed	J_V severed/ J_V intact
1	0.54	0.36	1.50	1.34
2	0.77	0.58	1.33	1.42
3	0.61	0.45	1.36	2.04
4	0.77	0.71	1.08	2.53
5	0.86	0.76	1.13	2.53
6	0.59	0.49	1.20	2.34
Mean ± SD	0.69 ± 0.12	0.56 ± 0.14	1.27 ± 0.14	2.03 ± 0.49
Range	0.54–0.86	0.36–0.76	1.08–1.50	1.34–2.53
	σ_{ABA} of *Helianthus annuus* L.			
1	0.95	0.44	2.16	1.21
2	0.83	0.70	1.18	1.14
3	0.91	0.44	2.08	1.72
4	0.84	0.61	1.39	1.52
5	0.84	0.43	1.96	2.20
6	0.94	0.62	1.52	3.11
7	0.97	0.78	1.26	2.14
8	0.95	0.77	1.24	1.34
Mean ± SD	0.91 ± 0.06	0.60 ± 0.14	1.60 ± 0.38	1.80 ± 0.62
Range	0.83–0.97	0.43–0.78	1.18–2.16	1.14–3.11

as well (Cruz et al. 1992; Stasovsky and Peterson 1993; North and Nobel 1995, 1996; Zimmermann and Steudle 1998).

The fact that ratios of J_{ABA} (−0.06 MPa)/J_{ABA} (−0.02 MPa) were always higher than those of J_V (−0.06 MPa)/J_V (−0.02 MPa) (Table 2), does not disagree with the above-mentioned conclusion that the endodermis of roots is an effective barrier to ABA. If there is a significant apoplastic barrier to the bypass flow of ABA in both species, there should be a considerable accumulation of ABA in front of and within the barrier, and J_{ABA} should increase as the water flow increases. On the other hand, using $(C_X^{ABA} + C_O^{ABA})/2$ as an estimate of the concentration of ABA within the barrier (Eqs. 1–3), should underestimate the real concentration in the permeation barrier. However, if the concentration in the barrier is underestimated, the real σ_{ABA} should be bigger than that estimated from C_X^{ABA}, C_O^{ABA}, J_V and J_{ABA} (see Eq. 2). This means that the 'apparent' values of the reflection coefficients of the roots given here represent a lower limit. The effect should, however, not be overestimated. The values of σ_{ABA} were, on average, somewhat smaller than values that were measured for salts and other polar solutes for maize and other species using the root pressure probe (e.g. Steudle and Frensch 1989; Steudle 1994). These latter measurements were performed at zero water flow and were therefore rather free of effects of concentration polarisations or of sweep-away effects. Effects of diffusional unstirred layers have been accounted for (Steudle and Frensch 1989). Hence, from these considerations we think that, at least for maize, the absolute value of σ_{ABA} should be smaller than unity.

Reflection coefficients were reduced at an increased suction force (water flow), but increased with increasing external ABA concentration. This supports the view that there was a substantial contribution by an apoplastic component of the J_{ABA} in the presence of low external concentrations which decreased as the external ABA concentration was raised. The reasons for the substantial changes in σ_{ABA} are not completely clear. They may be related to concentration polarisation effects (see above) which dominate at high concentrations. However, at low concentrations, the capacity of the apoplast of the permeation barrier in the root to take up ABA may be sufficient, which would tend to increase the relative contribution of apoplastic flow at low concentrations and reduce σ_{ABA}. At high concentrations, σ_{ABA} should be high. This shows that, for several reasons, root reflection coefficients measured in the presence of volume flows have to be treated with caution. This throws some doubt on literature data where root reflection coefficients have been measured at varying water flow and in the presence of high external solute concentrations (Schneider et al. 1997). The only way to circumvent the difficulties is to measure reflection coefficients in the absence of a water flow, e.g. by using the root pressure probe (Steudle 1994; Steudle and Peterson 1998). This, however, requires concentrations of the osmotic solute in the millimolar range, which is not applicable for ABA.

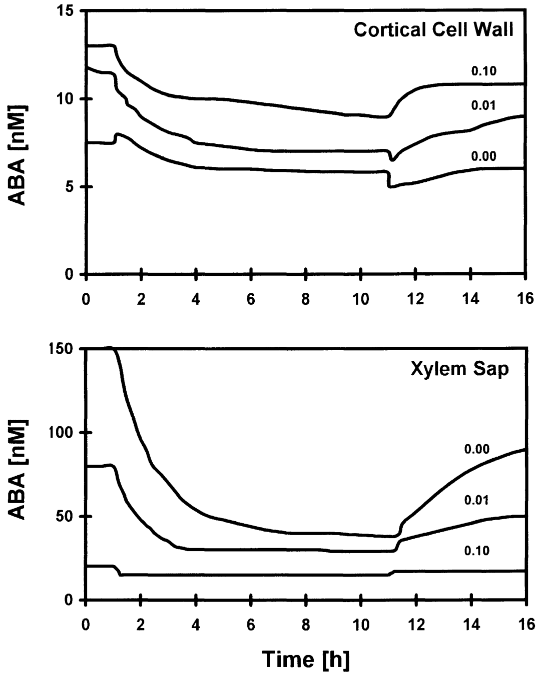

Fig. 8. Computer simulation of the time course of ABA concentration in cortical cell walls and the xylem saps of a model root. After 1 h, transpirational water flow was increased and after 11 h it was reduced again. The three curves show ABA changes when an apoplastic bypass-flow did not exist (*0.00*) or when either 1% (*0.01*) or 10% (*0.10*) of total ABA was transported apoplastically across the endodermis

We have used measured values of J_{ABA}, C_X^{ABA}, C_O^{ABA} and J_V to estimate the apparent reflection coefficient σ_{ABA} of maize and sunflower root systems for ABA. In order to do this, J_{ABA} was identified with the solvent-drag component only (see Eq. 2 of *Materials and methods*). The active ABA uptake (J^*_{ABA}) and its passive diffusion across the root cylinder $[P_S^{ABA} \cdot (C_O^{ABA} - C_X^{ABA})]$ were neglected. Values of permeability coefficients of ABA have been estimated for maize and sunflower by extrapolation to zero water flow (Eq. 1). The data (not shown) indicate that, for the species used, P_S^{ABA} was smaller by several orders of magnitude than the value given by Fiscus (1982) for bean (*Phaseolus vulgaris* L.) roots. Hence, the contribution of the diffusive component of J_{ABA} should have been small in the presence of the low gradients of ABA used in this paper (up to 500 nM). The active component can be neglected as well in the presence of hydrostatic pressure gradients and high rates of water flow. For sunflower, corn and runner bean roots, ABA uptake mediated by carrier systems only occurs in the very tips of primary roots and could not be detected in the root hair zone (Astle and Rubery 1983; Hartung and Dierich 1983).

At low external concentrations of ABA ($C_O^{ABA} <$ 10 nM), small amounts of ABA that are released symplastically to xylem vessels may cause artificially low reflection coefficients. When higher concentrations ($C_O^{ABA} >$ 20 nM) were used, this error appeared to be negligible and σ_{ABA} values for maize roots were in the range 0.5–0.7. This indicated a high bypass-flow of ABA in the cortical apoplast which should have a low reflection coefficient for this solute. In maize roots, dilution of ABA that is released from the symplast to the xylem is counteracted by solvent drag.

According to biophysical and anatomical studies, we conclude that the major barrier to radial solute flow is the endodermal layer (Steudle 1994; Steudle et al. 1993; Steudle and Peterson 1998). In maize, however, the endodermis was more permeable to ABA than in sunflower. Perhaps, this may be explained by a different chemical composition. As reported for *Clivia miniata* L. (Schreiber et al. 1994) and for five different monocotyledonous plants (Zeier and Schreiber 1998), the chemical composition of the endodermis may not be as hydrophobic as it is usually expected. The situation could be different with sunflower roots. Although no visible structures of the endodermis, such as Casparian bands or suberin deposits were detected microscopically, the apparent reflection coefficient (σ_{ABA}) was close to unity, indicating a low permeability of the endodermis to ABA. Despite the high σ_{ABA}, considerable amounts of ABA were transported apoplastically to the xylem vessels to leave the xylem ABA concentration unchanged or even slightly increased. This agrees well with predictions of a computer simulation, using the mathematical model of Slovik et al. (1995). According to the model, C_X^{ABA} was diluted when transpiration and water flow were increased, suggesting a predominantly symplastic transport. When an apoplastic bypass was incorporated into the Slovik model, dilution was counteracted significantly when 1% of ABA was dragged apoplastically to the xylem (Fig. 8).

Values of σ_{ABA} of 0.8–0.9 for sunflower roots are still sufficiently low to allow for an apoplastic bypass flow of ABA that compensates for dilution, at least to some extent. On the other hand, the high reflection coefficient of sunflower roots may explain the low ABA concentrations in the xylem sap of sunflower plants. When the permeability of the endodermis is low, ABA may be redistributed to the symplast at the endodermis. Another explanation of high σ_{ABA} is possible. Steudle and Brinckmann (1989) postulated that the hydraulic conductivity of individual root cortex cells of *Phaseolus coccineus* L. was larger by 2 orders of magnitude than that of entire root systems. This indicated a predominant cell-to-cell rather than an apoplastic transport of water and somewhat higher root reflection coefficients. Assuming similar conditions in sunflower, the symplastic transport component would be more pronounced resulting in a higher overall σ_{ABA} in sunflower as compared with maize.

When the pH of the medium was acidified down to the pK_a of ABA, the apparent reflection coefficient was significantly reduced in both roots of maize and sunflower. Two reasons may be important: (i) At low pH, the concentration of lipophilic, protonated ABAH was increased which should have reduced the reflection coefficient of cell membranes. (ii) The permeability of the endodermis could have been increased which should also result in a smaller σ_{ABA}. pH values of lower than 5 occur in the rhizosphere, for example, when roots use NH_4^+ as an N-source.

When root tips are cut off, an apoplastic bypass for ABA and water is opened which should result in a reduction of σ_{ABA} at an increased J_V (Steudle et al. 1993). As expected, a reduction of σ_{ABA} by severing was stronger in sunflower than in maize. At a first sight it may be astonishing that, after cutting, root reflection coefficients were still substantially larger than zero. However, this is theoretically expected. Although the hydraulic conductance was substantially increased by severing, it was still in the same range as that of the intact root. The result resembles findings with single roots of maize where puncturing of the root endodermis did not cause the reflection coefficient to assume values of zero (Steudle et al. 1993).

In conclusion, the results indicate a significant apoplastic bypass flow of ABA in roots of sunflower and maize and apparent root reflection coefficients which are smaller than unity. Reflection coefficients estimated from solvent drag of ABA varied considerably with water flow, indicating substantial effects of unstirred layers. When the water flow across the roots was increased, the rate of ABA translocated across the root cylinder increased. Flows of ABA also increased with increasing concentrations of ABA in the medium or when the pH of the medium was lowered. The data indicate an apoplastic bypass of ABA even in the root endodermis. There are remarkable differences between species which are consistent with the view that the endodermis is the main permeation barrier for ABA. The differences between the two species in σ_{ABA}, C_X^{ABA} and J_{ABA} should largely reflect differences in the transport properties of the endodermis. However, under natural conditions, transport properties of roots are determined by two tissue layers with Casparian bands. The hydroponically grown plants used in our experiments developed a Casparian band in the endodermal layer but were lacking it in the hypodermis. The existence of an exodermis may further change the root transport properties for ABA.

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